



DJI Spark mini-drone

The Spark will retail in India at Rs 43,000 that includes the drone, battery, a USB charger and three pairs of propellers. <http://dgit.in/DJIsparKIND>



Terminator is returning?

Arnold Schwarzenegger confirms that the Terminator franchise is making a comeback and he will be in the next film. <http://dgit.in/newTerminator>



From Cartography to GPS

Finding the right path has evolved to almost pinpoint level accuracy

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aps, in one form or another, have been crucial in the story of human civilisations.

Travelling vast

distances, prehistoric man communicated the details of the route back to the people left behind, and that is where mapping began in its true sense. Over the years, maps have evolved to a point where we can trust our smartphone navigation apps more than we would trust our own sense of direction. In fact, today everyone with a smartphone can become a cartographer with programs such as Google maps local guides. But this journey started long ago, when most of the roads in the world were unmapped.

TRACING THE HORIZON

While it is well known that the Egyptians had made maps and city plans on papyrus long ago, very little evidence of the same made it to our time due to the fragile nature of the medium. The earliest known instance is more of a town plan that was discovered in the 1960s. The wall painting discovered there by James Mellaart was carbon

dated to 6200 BCE and it appears to represent a crude layout of the town and the nearby stratovolcano Mount Hasan. While there is speculation whether this is a map at all to begin with, this serves as an interesting example of what mapping was all about back then – subjective interpretation.

Another helpful instance is a Babylonian map from about 600 BC. In the representation, Babylon was represented at the centre of the map, which was a common practice of the time, while several other cities like Assyria and Urartu were represented around it.

A SCIENCE

It was the Greeks who pushed cartography from a subjective artform to a science necessary for the society. Eminent scholars like Ptolemy, Herodotus, Anaximander and Eratosthenes had

significant impact on the field of geography. Anaximander, for instance, was the first to draw a map of the known world, while Pythagoras of Samos speculated about a spherical earth with a central fire at its core. These maps were made and improved over the years. Based on Anaximander's map, Hecataeus of Miletus created a new, more expansive map.

As the next major upgrade to mapping techniques, Eratosthenes incorporated parallels and meridians in 2nd Century BC, which indicated his understanding of the spherical earth theory, something some of us are apparently still struggling with. Around a century later, Posidonius, in his work "about the ocean and the adjacent areas" calculated earth's circumference at 240,000 stadia, close to the actual value of 24,901 miles.

A TOOL

While Ptolemy's original maps were never found, he had described, in his work, a system of using latitudes and longitudes as well as astronomical observations from particular areas to develop better maps. His work was informative enough for cartographers of 1300 AD to recreate the Ptolemy map. This was the first time cartography was a numerical science.

Over the next few centuries, the mapping technique didn't change much. However, maps were increasingly detailed as the known world was explored further. The next big cartographical milestone came with the popularisation of printing.

Around the 16th century, tools like the pendulum clock, telescopes, as well as established theoretical tools like the logarithmic tables, calculus, law of gravity helped scientists make significant progress. A major improvement came through the publication of Mercator's maps. These maps were the very first to take into account the spherical-to-flat conversion that needs to happen when representing earth on a flat surface. As a result, these became crucial tools for naval navigation be-